

Aluno(a):	Data:
Professor(a):	Nota:

Orientações

Para cada função lógica:

- Monte o Mapa de Karnaugh;
- Preencha corretamente as células;
- Faça os maiores agrupamentos possíveis;
- Determine a expressão lógica simplificada.

1. Mapa de Karnaugh com 2 variáveis

(J)

Simplifique as funções utilizando mapas de Karnaugh com as variáveis A e B .

$$F(A, B) = \Sigma m(0, 1, 2, 3)$$

(A)

$$F(A, B) = \Sigma m(0, 1)$$

(B)

$$F(A, B) = \Sigma m(0, 2)$$

(C)

$$F(A, B) = \Sigma m(1, 3)$$

(D)

$$F(A, B) = \Sigma m(2, 3)$$

(E)

$$F(A, B) = \Sigma m(0, 3)$$

(F)

$$F(A, B) = \Sigma m(1, 2)$$

(G)

$$F(A, B) = \Sigma m(0, 1, 2)$$

(H)

$$F(A, B) = \Sigma m(0, 1, 3)$$

(I)

$$F(A, B) = \Sigma m(0, 2, 3)$$

2. Mapa de Karnaugh com 3 variáveis

Simplifique as funções utilizando mapas de Karnaugh com as variáveis A , B e C .

(A)

$$F(A, B, C) = \Sigma m(0, 1)$$

(B)

$$F(A, B, C) = \Sigma m(2, 3)$$

(C)

$$F(A, B, C) = \Sigma m(0, 2, 4, 6)$$

(D)

$$F(A, B, C) = \Sigma m(1, 3, 5, 7)$$

(E)

$$F(A, B, C) = \Sigma m(0, 1, 2, 3)$$

(F)

$$F(A, B, C) = \Sigma m(4, 5, 6, 7)$$

(G)

$$F(A, B, C) = \Sigma m(0, 1, 4, 5)$$

(H)

$$F(A, B, C) = \Sigma m(2, 3, 6, 7)$$

$$(I) \quad F(A, B, C) = \Sigma m(0, 1, 2, 3, 5, 7)$$

$$(J) \quad F(A, B, C) = \Sigma m(1, 2, 3, 5, 7)$$

(F)

$$F(A, B, C, D) = \Sigma m(2, 3, 6, 7, 10, 11, 14, 15)$$

(G)

$$F(A, B, C, D) = \Sigma m(0, 1, 2, 3, 8, 9, 10, 11)$$

3. Mapa de Karnaugh com 4 variáveis

Simplifique as funções utilizando mapas de Karnaugh com as variáveis A , B , C e D .

(H)

$$(A) \quad F(A, B, C, D) = \Sigma m(0, 1)$$

$$F(A, B, C, D) = \Sigma m(4, 5, 6, 7, 12, 13, 14, 15)$$

$$(B) \quad F(A, B, C, D) = \Sigma m(0, 1, 2, 3)$$

(I)

$$(C) \quad F(A, B, C, D) = \Sigma m(0, 2, 4, 6)$$

$$F(A, B, C, D) = \Sigma m(0, 2, 3, 5, 7, 8, 10, 11, 13, 15)$$

$$(D) \quad F(A, B, C, D) = \Sigma m(1, 3, 5, 7)$$

(J)

$$(E) \quad F(A, B, C, D) = \Sigma m(0, 1, 4, 5, 8, 9, 12, 13)$$

$$F(A, B, C, D) = \Sigma m(0, 2, 3, 5, 7, 8, 10, 11, 13, 15)$$